



Introduction to Macroeconomics

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Solution: Problem Set 1 2025

1 Introduction

Solution 1 (Exam August 2013)

Which of the following statements are correct ?

- (I) GDP and the wealth of a person are stock variables.
 - (II) According to monetarists monetary policy is inefficient if it is not anticipated.
 - (III) A correlation between two variables, say x and y , always implies that there is a causal relation between those two. The goal of an economist is to put forward hypotheses about this causal link and to test them using available data.
 - (IV) In an economic model endogenous variables are explained by exogenous variables.
- (a) Only IV is correct.
 - (b) Only I is false.
 - (c) II and III are correct and the rest is false.
 - (d) CII and IV are correct and the rest is false.

Answer (a)

2 Gross domestic product

Solution 2 (Exam June 2014)

Answer (d)

Country A : We have information on the GDP at market prices and we need to find the National Income (NI). We know the following accounting identities :

$$NI = GDP \text{ at market price} + NFIA - \text{indirect taxes} + \text{subsidies} - \text{capital depreciation}$$

$$NI = 110 + 5 - \text{indirect taxes} + \text{subsidies} - (\text{subsidies} - \text{indirect taxes}) = 115$$

Country B : We have the information on National Income (NI) and we wish to find the GDP at market prices. The above identity can be rewritten as :

$$GDP \text{ at market price} = NI + \text{capital depreciation} - \text{subsidies} + \text{indirect taxes} - NFIA = 95$$

Since there are only 2 countries in the world economy, it must be that $NFIA_B = -NFIA_A$.

Solution 3

Part 1 :

Nominal GDP values the production of goods and services at current prices. Therefore, the production of two different years (A and B) can be the same in terms of quantities, but as the value of goods and services is at current prices, if the price has risen between A and B, the nominal GDP for B will be higher than for A. In other words, a rise in nominal GDP can thus be attributed to a rise in prices, a rise in quantities or to both. If there has been no growth and the quantities produced are the same, a rise in prices will induce a rise in nominal GDP. Therefore, the evolution of nominal GDP across years is not a good indicator of economic growth.

A better indicator of economic growth would be the evolution of real GDP, in which the production is valued at constant prices (at the price of the year of reference, 20YY). Therefore, a growth in real GDP between two years does indicate a growth in the quantities produced. If we have the nominal GDP for year 20XX ($nGDP_{20XX}$) and we are interested in the real GDP ($rGDP_{20XX}$) for that year, we need the GDP deflator for year 20XX with base 100 in 20YY ($GDPdeflator_{20XX}$) :

$$rGDP_{20XX} = \frac{nGDP_{20XX}}{GDPdeflator_{20XX}} \cdot 100$$

This relationship between the nominal and the real GDP allows us to derive the relationship between the percentage change in nominal GDP and the percentage change in real GDP. We first apply natural logarithms on both sides of the equation :

$$\ln(rGDP_{20XX}) = \ln\left(\frac{nGDP_{20XX}}{GDPdeflator_{20XX}} \cdot 100\right)$$

$$\ln(rGDP_{20XX}) = \ln(nGDP_{20XX}) - \ln(GDPdeflator_{20XX}) + \ln(100)$$

Now we take the first derivative with respect to time on both sides of the equation, knowing that the derivative of $\ln(100)$, which is a number, with respect to time is equal to 0 :

$$\frac{\partial \ln(rGDP_{20XX})}{\partial t} = \frac{\partial [\ln(nGDP_{20XX}) - \ln(GDPdeflator_{20XX}) + \ln(100)]}{\partial t}$$

$$\frac{\partial \ln(rGDP_{20XX})}{\partial t} = \frac{\partial \ln(nGDP_{20XX})}{\partial t} - \frac{\partial \ln(GDPdeflator_{20XX})}{\partial t} + \frac{\partial \ln(100)}{\partial t}$$

$$\frac{\partial \ln(rGDP_{20XX})}{\partial(rGDP_{20XX})} \frac{\partial(rGDP_{20XX})}{\partial t} = \frac{\partial \ln(nGDP_{20XX})}{\partial(nGDP_{20XX})} \frac{\partial(nGDP_{20XX})}{\partial t} - \frac{\partial \ln(GDPdeflator_{20XX})}{\partial(GDPdeflator_{20XX})} \frac{\partial(GDPdeflator_{20XX})}{\partial t}$$

$$\frac{\frac{\partial(rGDP_{20XX})}{\partial t}}{rGDP_{20XX}} = \frac{\frac{\partial(nGDP_{20XX})}{\partial t}}{nGDP_{20XX}} - \frac{\frac{\partial(GDPdeflator_{20XX})}{\partial t}}{GDPdeflator_{20XX}}$$

If we take $\frac{\frac{\partial(rGDP_{20XX})}{\partial t}}{rGDP_{20XX}}$, we see that it corresponds to the change in $rGDP_{20XX}$ relative to the total value.

Therefore, the relative change in the real GDP is equal to the relative change in nominal GDP minus the relative change in the GDP deflator. If we transform the expression we have obtained from its continuous form (with derivatives) to its discrete form (annual changes), which is an approximation, and we multiply both sides of the equation by 100, to obtain percentage changes over a given period of time, we obtain :

$$\%_change_in_real_GDP = \%_change_in_nominal_GDP - \%_change_in_GDP_deflator$$

Part 2 :

In this problem, the percentage change in nominal GDP is 12%, and the percentage change in GDP deflator is $\frac{108-100}{100} \cdot 100\% = 8\%$. Therefore, using the expression we derived in part 1 :

$$\%_change_in_real_GDP = 12\% - 8\% = 4\%$$

Solution 4

Part 1 :

Let's call $P_{m,X}$ and $P_{h,X}$ the price of milk and the price of honey on year X , respectively. Let's call $Q_{m,Y}$ and $Q_{h,Y}$ the quantity of milk and the quantity of honey on year Y , respectively.

We first calculate the nominal GDP :

$$nGDP_{2003} = P_{m,2003} \cdot Q_{m,2003} + P_{h,2003} \cdot Q_{h,2003} = 1 \cdot 100 + 2 \cdot 50 = 200$$

$$nGDP_{2004} = P_{m,2004} \cdot Q_{m,2004} + P_{h,2004} \cdot Q_{h,2004} = 1 \cdot 200 + 2 \cdot 100 = 400$$

$$nGDP_{2005} = P_{m,2005} \cdot Q_{m,2005} + P_{h,2005} \cdot Q_{h,2005} = 2 \cdot 200 + 4 \cdot 100 = 800$$

Then the real GDP for each year using 2003 as the base year. This means that, for year Y, we use the quantities that correspond to year Y, but valued at the 2003 price :

$$rGDP_{2003} = P_{m,2003} \cdot Q_{m,2003} + P_{h,2003} \cdot Q_{h,2003} = 1 \cdot 100 + 2 \cdot 50 = 200$$

$$rGDP_{2004} = P_{m,2003} \cdot Q_{m,2004} + P_{h,2003} \cdot Q_{h,2004} = 1 \cdot 200 + 2 \cdot 100 = 400$$

$$rGDP_{2005} = P_{m,2003} \cdot Q_{m,2005} + P_{h,2003} \cdot Q_{h,2005} = 1 \cdot 200 + 2 \cdot 100 = 400$$

To calculate the GDP deflator, we use :

$$GDPdeflator_Y = \frac{nGDP_Y}{rGDP_Y} \cdot 100$$

Therefore :

$$GDPdeflator_{2003} = \frac{nGDP_{2003}}{rGDP_{2003}} \cdot 100 = \frac{200}{200} \cdot 100 = 100$$

$$GDPdeflator_{2004} = \frac{nGDP_{2004}}{rGDP_{2004}} \cdot 100 = \frac{400}{400} \cdot 100 = 100$$

$$GDPdeflator_{2005} = \frac{nGDP_{2005}}{rGDP_{2005}} \cdot 100 = \frac{800}{400} \cdot 100 = 200$$

Part 2 :

We are asked to calculate the percentage change in nominal GDP, real GDP and the GDP deflator in 2004 and 2005 from the preceding year :

2004 :

$$\%change \text{ in } nGDP = \frac{nGDP_{2004} - nGDP_{2003}}{nGDP_{2003}} \cdot 100\% = \frac{400 - 200}{200} \cdot 100\% = 100\%$$

$$\%change \text{ in } rGDP = \frac{rGDP_{2004} - rGDP_{2003}}{rGDP_{2003}} \cdot 100\% = \frac{400 - 200}{200} \cdot 100\% = 100\%$$

$$\%change \text{ in } GDPdefl = \frac{GDPdefl_{2004} - GDPdefl_{2003}}{GDPdefl_{2003}} \cdot 100\% = \frac{100 - 100}{100} \cdot 100\% = 0\%$$

From 2003 to 2004 we observe that the change in nominal GDP is entirely due to the change in the real GDP : the %change in the nominal GDP is equal to the %change in the real GDP, while there is no change in the GDP deflator (and thus in the price level). If we look at the data in the table, we see that between 2003 and 2004 the prices of milk and honey stay the same, while production of each good doubles. The doubling of the production level of each good explains the 100% change in the real GDP and in the nominal GDP.

2005 :

$$\%change\ in\ nGDP = \frac{nGDP_{2005} - nGDP_{2004}}{nGDP_{2004}} \cdot 100\% = \frac{800 - 400}{400} \cdot 100\% = 100\%$$

$$\%change\ in\ rGDP = \frac{rGDP_{2005} - rGDP_{2004}}{rGDP_{2004}} \cdot 100\% = \frac{400 - 400}{400} \cdot 100\% = 0\%$$

$$\%change\ in\ GDPdefl = \frac{GDPdefl_{2005} - GDPdefl_{2004}}{GDPdefl_{2004}} \cdot 100\% = \frac{200 - 100}{100} \cdot 100\% = 100\%$$

From 2004 to 2005 we observe that the change in nominal GDP is entirely due to the change in the price level : the %change in the nominal GDP is equal to the %change in GDP deflator, while there is no change in the rGDP. If we look at the data in the table, we see that between 2004 and 2005 the prices of milk and honey double, while production of each good stays the same. The doubling of the price of each good explains the 100% change in the GDP deflator and thus in the nominal GDP.

Part 3 :

Economic well-being increases when the production of goods and services increases. In 2004 and 2005 the nominal GDP increases by 100% with respect to the preceding year. However, in 2004 the prices stayed the same, while the production of each good doubled, while in 2005 the prices doubled but the quantities of each good produced remained the same, as can be seen on the table. Therefore, percentage increase in nominal GDP is not a good indicator of the increase in economic well-being, as while economic well-being increased in 2004, it did not in 2005.

In problem 2 we derived the relationship between the percentage changes in nominal GDP, real GDP and the GDP deflator :

$$\%_change_in_nominal_GDP = \%_change_in_real_GDP + \%_change_in_GDP_deflator$$

We see from the results in part 2 that this expression is fulfilled. In part 2 we also saw that the increase in the nominal GDP for 2004 was entirely due to the increase in the real GDP, while for 2005 it was entirely due to the increase in the GDP deflator (and thus in the price level).

We can conclude that the increase in real GDP is a much better indicator of the change in economic well-being, as it only changes if there are changes in quantities. In its calculation,

we always use the prices of the base year, as seen in part 1, and therefore the real GDP is not affected by changes in prices that happen along the years.

Solution 5

This table is the Leontieff table, which represents the economy of a country and summarizes the three approaches available to economists to calculate the GDP :

1. the production approach (or value-added approach) : $GDP = \text{total output} - \text{total purchase of intermediate goods}$. Why? Because the GDP only includes the value of the final goods and services and, as a result, we must subtract the intermediate goods from the total output. Subtracting the intermediate goods from the total output allows avoiding double counting the value generated in the production of these intermediate goods and services.

2. the income approach : $GDP = \text{labor income} + \text{capital income} + \text{profits}$. The rationale behind this formula is that all the value added that results from the production process is shared between the production factors or inputs, which are labor and capital, and once these inputs have been remunerated, the rest goes to the entrepreneur as profits.

3. the expenditure approach : $\text{Final net demand} = \text{Consumption} + \text{Investment} + \text{eXports} - \text{iMports} = C + I + X - M$. The rationale is that all the final goods and services that are produced (that is, the final output) are bought by somebody. In a closed economy without government, the final net demand is equal to consumption plus investment. Why? Because all the final goods and services are either used for private consumption or are used to make an investment. However, in an open economy without government, it is possible that production exceeds the domestic consumption $C + I$, which can only be consistent with positive net exports ($X - M > 0$), or that production is below $C + I$, which is only consistent with positive net imports (or negative net exports, $X - M < 0$). It should be noted that $X - M$ is called the trade balance.

The top left of the table is used to determine the total sales of intermediate goods (and services), which must be equal to the total purchase of intermediate goods. How do we read the table? The Manufacturing sector, for example, sells (manufactured) intermediate goods to the Agriculture sector for a market value of 2, to the Manufacturing sector for a value of 10 and to the Services sector for a value of 4. Therefore, it sells a total value of $2 + 10 + 4 = 16$ of intermediate goods, which is what appears on the column "Total sales of intermediate goods". The Agriculture Sector sells intermediate goods for a value of 1 to the same sector, the Manufacturing sector sells intermediate goods (and services) for a value of 2 to the Agriculture Sector and the Services Sector sells intermediate goods (or more precisely services) for a value 1 to the Agriculture Sector. Therefore, the total purchase of intermediate goods (and services) by the Agriculture Sector is $1 + 2 + 1 = 4$, which is what appears on the row "Total purchases of Intermediate goods". **Therefore, A is the value of the sales of the Agriculture sector to the Manufacturing sector.**

Therefore, the **total purchase of intermediate goods (and services)** is $24 + A$. Now, in order to be able to calculate the GDP using the production approach, we need to determine total output. It is calculated at the bottom left of the table, last row, and it is equal to **62**. As a result, using the **production approach** :

$$GDP_{production_approach} = total_output - total_purchase_of_intermediate_goods = 62 - (24 + A) = 38 - A$$

The bottom left of the table is also used to determine the GDP using the **income approach**. For example, labor is remunerated for a value of 5 in the Agriculture sector, for a value of 7 in the Manufacturing sector and for a value of 4 in the Services Sector. In the economy, the total labor income is thus $5 + 7 + 4 = 16$. If we adopt the same approach for the Capital income, we find that it is 12 and, finally, the total profits are 7. Therefore, using the **income approach** :

$$GDP_{income_approach} = labor_income + capital_income + profits = 16 + 12 + 7 = 35$$

Finally, the top right top of the table is used to determine the GDP using the expenditure approach. For example, the value of agricultural goods (and services) used for consumption is 6, the value of agricultural goods used for investment is 1, the value of agricultural goods exported is 4 and the value of agricultural goods (and services) imported is 3. As a result, the final net demand of agricultural goods is equal to $6 + 1 + 4 - 3 = 8$. If we do the same for the manufactured goods and the services we obtain the final net demand of manufactured goods, which is 18, and the final net demand of services, which is 9, respectively. Therefore, using the **expenditure approach** :

$$\begin{aligned} GDP_{expenditure_approach} &= final_net_demand_of_agriculture_goods \\ &\quad + final_net_demand_of_manufacturing_goods \\ &\quad + final_net_demand_of_services \\ &= 8 + 18 + 9 = 35 \end{aligned}$$

On the other hand, total consumption is equal to the sum of the consumption of agriculture goods, manufactured goods and services), which is equal to $6 + 10 + 6 = 22$. We can also calculate total investment (10), total exports (14) and total imports (11), and then calculate the GDP using the expenditure approach as defined at the beginning of the solution to this exercise :

$$GDP_{expenditure_approach} = C + I + X - M = 22 + 10 + 14 - 11 = 35$$

Since :

$$GDP_{production_approach} = GDP_{income_approach} = GDP_{expenditure_approach} = 35$$

and we had found that $GDP_{production_approach} = 38 - A$, it is now possible to calculate the value of A :

$$\begin{aligned}GDP_{production_approach} &= 38 - A = 35 \\A &= 38 - 35 = 3\end{aligned}$$

Therefore, we have provided the answers for questions 1 and 2 :the Leontieff table,allows calculating the GDP using the three approaches described above. A is the value of the sales of the Agriculture sector to the Manufacturing sector and it is equal to 3. The GDP has been calculated using the three approaches, and it is equal to 35.

Solution 6

1. The value of GDP has increased. There are now additional workers in the formal sector of the economy that were previously not accounted for in the calculation of GDP.
2. The value of a welfare indicator that takes into account household work and leisure time would decrease. This is because the entry of women into formal labour market would decrease the time spent on leisure and household work.
3. (This is an open question, and any reasonable argument would be accepted as an answer. Here we list some examples.) The increase of female participation in the labour force could lead to an improvement in the economic well-being, for instance through reduction in inequalities or reduction in risk of poverty or vulnerability of women following a divorce. On the other hand, there are examples in which female participation in labour force could lead to a decrease in economic well-being, if it is found that parental absenteeism has negative effect on child's health or education, and vice versa. In general, the question of what constitutes a well-being of an individual, and therefore a well-being of a society, is subject to an ongoing debate not only in economics, but across other fields as well. Therefore, seeing the normative nature of such a measure, it would be difficult to form one indicator which encompasses all aspects of an economic well-being of a nation. In addition, some aspects (like happiness levels, or impact on climate change) are difficult to measure. Also, it would be difficult to aggregate all factors into one single indicator (seeing as the decision on how much weight to put on each indicator is subjective). This is why GDP has become the most widely used measure of economic progress and well-being, as it is easily measured and comparable across countries, and is strongly correlated with important indicators of well-being, such as child mortality, life expectancy, happiness, etc.

3 Additional exercises (will not be discussed in the seminar)

Solution 7

⇒ **Answer (d)** GDP definition : "total market value of all final goods and services produced within a country in a given period of time."

I. True : it does not include intermediate goods and services, as the value of these is already included in the final goods and services. If we added to the value of the final goods and services the value of the intermediate ones, we would be counting these twice in the GDP.

II : False : the production of final goods and services of foreign firms that operate within the country is also included in the GDP.

III : False : it is not the totality of the national output, but the final output (without the intermediate goods and services). Moreover, it is not correct to use the term "national output", as the final goods and services produced within the national territory by foreign firms is also included in the GDP.

IV : True : it includes the goods and services currently produced in the national territory, and not the transactions involving goods produced in the past (if some good produced in the previous period is exchanged in the market on this period, it does not enter the GDP).

Solution 8

$$GDP = C + I + G + X - N$$

$$GNP = GDP + NFIA$$

$$GNP_{factor\ prices} = GNP - indirect\ taxes + subsidies$$

$$NI = GNP_{factor\ prices} - depreciation$$

- (a) Per definition $GNP = GDP + NFIA$. We do not have data on NFIA for country A in the table, but it is assumed in this exercise that there are only two countries. Thus $NFIA_A = -NFIA_B$ i.e. $NFIA_A = -5$. GNP in country A is lower than its GDP by 5. (a) is false
- (b) This resembles the definition of GNP (except for the fact that to calculate GNP we would also subtract income received by foreign firms in the domestic market). It is not the definition of GDP calculated using the expenditure approach.
- (c)

$$GDP_A = 60 + 30 + 40 + 10 - 15 = 125$$

$$GNP_A = 125 - 5 = 120$$

$$GNP_{A\ factor\ prices} = 120 - 5 + 10 = 125$$

$$NI_A = 125 - 10 = 115$$

It is higher than $NI_B = 40$. So (c) is **correct**

(d)

Option (d) is wrong. We know that the difference between national income (which is always at factor prices) and the GNP at factor prices is that we subtract the capital depreciation of the latter to find the former. Since capital depreciation must be either null or positive, it must be that the GNP at factor prices is always equal or bigger than the national income (which is given to be 40 in the exercise).